

JAVA ASSIGNMENT - 1

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Subject : Programming in Java .

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1. Method overloading:

If two or more methods in a class has same name but different parameter list, it is known as method overloading. Java implement static polymorphism using method overloading can be done by changing number of arguments (or) changing the data types of argument. If two (or) more methods have same name and same parameters list, but differs in return type are not said to be overload method.

eg:

```
public class methodoverloading {  
    public int Addition (int a, int b) {  
        System.out.println ("method 1 is called")  
        return a + b;  
    }  
    public int Addition (int a, int b, int c) {  
        System.out.println ("method 2 is called")  
        return a + b + c;  
    }  
    public static void main (String [] args) {  
        int Answer1 = Addition (5, 6);  
        System.out.println ("Answer1=" + Answer1);  
        System.out.println (".....");  
        int Answer2 = Addition (5, 6, 7);  
        System.out.println ("Answer2=" + Answer2);  
    }  
}
```

Output :

method 1 is called

Answer 1 = 11

- - - - -

method 2 is called

Answer 2 = 18

2. Inheritance in java :

In java, inheritance is a mechanism which one object can get all the properties and behaviour of a parent object. In inheritance there is a relationship between two classes.

Eg: Bike, Car, bus are Vehicle
Child of Vehicle are bike, Car, bus
Parent class: Vehicle.

Types :

- * Single inheritance
- * Multiple inheritance
- * Multi-level inheritance.
- * Hierarchical inheritance
- * Hybrid inheritance.

Single inheritance :

Single inheritance extends another, only one class (parent class).

eg: Animal
 ↓
 Cat.

Multi-level inheritance :

A class extends a class and that class extends another class.

eg: Animal
 ↓
 Cat
 ↓
 Dog.

Hierarchical inheritance :

When a class is inherited by one (or) more subclass.

eg: Animal
 / \
 Dog Cat

Hybrid inheritance :

Hybrid inheritance is a type of inheritance that combines two (or) more types of inheritance such as single, multi-level in one program.

3. Overriding:

When a method in a sub class has same name and same method signature as a method in its superclass, then the method is known as overridden method.

eg :

```
class Animal {  
    public void eat () {  
        System.out.println ("Generic Animal eating");  
    }  
}  
class Dog extends Animal {  
    public void eat () {  
        System.out.println ("Dog extends eats meat");  
    }  
    public static void main (String [] args) {  
        Animal a = new Dog ();  
        a.eat ();  
    }  
}
```

Output :

Dog eats meat.

4. Access Control:

Encapsulation provides an important attribute which is called access control. Through encapsulation we can control what parts of a program can access the members of a class.

Types:

- i) public
- ii) private
- iii) protected
- iv) Default.

Public:

When a member of a class is specified as public, then that member can be accessed by other code. The public access modifier is specified using the keyword "public". There is no restriction on the scope of public data members.

private:

When a member of a class is specified as private, then the member can be accessed by other member of its class. Keyword - "private".

Protected :

The methods and data members declared as protected are accessible within the same package (or) subclass in different packages. Its keyword is protected.

Default :

When no access modifier is specified for a class, method (or) data member, it is said to have the access modifier by default. This means only classes within the same package can access it.